

Integration — Lesson 20 Test: Capstone, Putting It All Together

10 questions mixing the whole course · show your work in the box · calculators allowed · answers need UNITS where they apply · use $g = 10 \text{ m/s}^2$ and radians

Name: _____ Date: _____ Score: _____ / 10

- 1.** (adding rate $\times$ time pieces) You bike at 10 mph for 2 hours, then at 14 mph for 1.5 hours. 1 pt
What is your total distance?

- 2.** (area under a graph) A train's speed rises in a straight line from 4 m/s to 10 m/s over 6 seconds. Use geometry (a trapezoid, or a rectangle plus a triangle) to find the distance traveled. 1 pt

- 3.** (Fundamental Theorem) Evaluate $\int$ from 1 to 3 of $(2x + 1) dx$. 1 pt

4. (signed area: net vs total) A cart's velocity is $v(t) = 4 - 2t$ m/s for $0 \leq t \leq 4$. Find (a) the net displacement and (b) the total distance traveled. 1 pt

5. (area between curves) Find the area of the region trapped between $y = x$ and $y = x^2$ (they cross at $x = 0$ and $x = 1$). 1 pt

6. (gravity, $g = 10$) A ball is thrown straight up from the ground at 30 m/s. When does it reach its highest point, and how high is that? 1 pt

7. (average value) Find the average value of $f(x) = x^2$ on the interval $[0, 3]$. 1 pt

8. (e^x and $1/x$) Evaluate both: (a) $\int$ from 0 to 1 of $e^x dx$ (exact, then round to two decimals; $e \approx 2.718$) and (b) $\int$ from 1 to e of $(2/x) dx$.

1 pt

9. (linear inside) Evaluate $\int$ from 0 to $\pi/4$ of $\cos(2x) dx$.

1 pt

10. (u-substitution) Evaluate $\int$ from 0 to 1 of $2x(x^2 + 1)^3 dx$.

1 pt
